

Financial Intermediation and Delegated Monitoring

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ABSTRACT

Diamond asks why financial intermediaries exist when borrowers could in principle contract directly with many small lenders. He builds a model in which an entrepreneur's project return is privately observed, so lenders must either incur a deadweight bankruptcy penalty under a standard debt contract or pay a fixed cost K to monitor the outcome directly. With m lenders, direct monitoring costs mK in aggregate, which is prohibitive or invites free-riding as m grows. Delegating monitoring to a single intermediary solves the duplication problem but creates a second incentive problem: the intermediary itself is not monitored by its depositors. The paper's central result is that diversification resolves this second layer of agency cost even though every agent in the economy is risk neutral. As the intermediary lends to an increasing number of borrowers whose project returns are independent, the probability that pooled repayments fall short of the amount owed to depositors shrinks toward zero, so the expected cost of providing the intermediary with incentives approaches zero as well. The optimal contract at both layers is shown to be riskless-illusion debt: a fixed claim with a non-pecuniary bankruptcy penalty for shortfall. The analysis is extended to risk-averse agents and reconciled with the Leland-Pyle signalling model of intermediation, correcting an error in that paper's central proposition along the way.

01. Introduction

Diamond (1984) is one of the foundational papers of the modern theory of financial intermediation, alongside Leland and Pyle (1977) and Diamond and Dybvig (1983), the latter already reviewed in this library. Where Diamond-Dybvig explains why banks are fragile – why a bank financing illiquid assets with liquid demand deposits is exposed to self-fulfilling runs – the present paper explains why banks (and financial intermediaries generally) exist in the first place, independent of any liquidity or run consideration. The two papers were written by the same author within roughly a year of each other and are best read as a pair: this one supplies the reason an intermediary is a Pareto improvement over direct lending; Diamond-Dybvig then shows that the very feature that makes intermediation valuable – the intermediary holding illiquid, privately-monitored assets funded with liquid claims – is also what makes it fragile.

The question the paper poses is deceptively simple: if information about a borrower's ability to repay is costly to produce, and many small lenders each need that information, why does the economy not simply have each lender either monitor directly or free-ride on others' monitoring? Why insert a specialized intermediary, funded by deposits, between ultimate savers and ultimate borrowers? Diamond's answer is that the intermediary economizes on the total resources spent monitoring, but only if the intermediary itself can be trusted with depositors' money without being monitored in turn – and diversification, not reputation, scale economies, or superior monitoring technology, is what allows the intermediary to be trusted at a vanishing cost.

02. Research Problem and Hypothesis

2.1 Research Question

The paper asks whether – and under what conditions – delegating the monitoring of privately-informed borrowers to a single financial intermediary funded by many depositors can dominate both (i) direct lending under costly-state-verification-style debt contracts with deadweight bankruptcy penalties, and (ii) direct monitoring by each individual lender. More precisely: does the delegation of monitoring to an intermediary itself generate a net incentive cost (a "cost of delegation") large enough to offset the gross saving from avoiding duplicated monitoring, and if so, what determines the size of that delegation cost.

2.2 Motivation

By 1984 the finance literature had several models explaining a gross benefit from delegating information production to intermediaries (Leland and Pyle (1977); Campbell (1979); Chan (1983)), and a well-developed single-agent, single-principal contracting literature (Harris and Raviv (1979); Holmstrom (1979); Shavell (1979)) explaining when monitoring resolves moral hazard between one lender and one borrower. Neither strand closed the loop: none of these papers netted the cost of

controlling the intermediary's own incentive problem against the gross benefit of delegation. A theory of intermediation that ignores the agency problem between the intermediary and its own depositors is incomplete, because the intermediary that receives unobserved payments from borrowers has exactly the same temptation to under-report that the borrower had in the first place. Absent an answer to this second-layer problem, the case for intermediaries existing at all was arguably weaker than the case against them (an additional layer of unmonitored agents standing between savers and investment).

2.3 Existing Literature

Two branches feed directly into the model. The single-principal agency literature (Harris-Raviv, Holmstrom, Shavell) established that costly monitoring can restore first-best-like incentives between one principal and one agent, but did not consider what happens when the "principal" role is itself distributed across many small lenders who cannot coordinate monitoring. Townsend (1979) had already shown, in the costly state verification framework, that debt with (possibly stochastic) verification is the cost-minimizing contract when only the borrower observes the outcome for free; Diamond's Proposition 1 is the deterministic-verification special case of this result and supplies the microfoundation for why the underlying loan contract, and ultimately the deposit contract, take the form of debt. On the intermediation side, Leland and Pyle (1977) modelled ex-ante adverse selection (entrepreneurs know more about expected project quality before financing) and showed that retained equity is a costly signal; they speculated, without formal analysis, that intermediaries might collect this information at lower cost, but did not check whether the intermediary's own incentive problem eats the saving.

2.4 Research Gap

What was missing was a model of ex-post information asymmetry (the lender cannot verify realized project returns without cost) combined with an explicit accounting of the intermediary's own incentive-compatibility constraint vis-à-vis its depositors, in an environment with many borrowers so that the diversification margin can be studied formally. No existing paper asked when the sum of (a) the intermediary's physical monitoring cost and (b) its own delegation cost is lower than the cost of direct monitoring or direct (unmonitored) debt contracting.

2.5 Contribution

PAPER The paper's claimed contribution is threefold: (i) a full characterization of the optimal direct debt contract with non-pecuniary bankruptcy penalties under ex-post asymmetric information (Proposition 1, essentially a two-line special case of Townsend (1979) and Gale-Hellwig (1985) later formalise more generally); (ii) a formal proof that the per-borrower cost of delegating monitoring to an intermediary vanishes as the number of independent borrowers grows (Proposition 2), so that diversification – not risk sharing, not superior monitoring technology – is the source of intermediation's net benefit; and (iii) an extension of the diversification argument to risk-averse agents that distinguishes "diversification by subdividing risk" from "diversification by adding risk" (a distinction tied to Samuelson's 1963 "fallacy of large numbers"), together with a correction of an algebraic error in Leland and Pyle's own signalling model. **CRITIQUE** The claimed contribution

is largely delivered for (i) and (ii); (iii) is more limited in practice than the paper's framing suggests, because the risk-averse extension explicitly abandons the wealth constraint that does all the work in the main (risk-neutral) result, so it functions as a robustness check on the source of the effect rather than as an independently load-bearing result (see Section 5).

03. Theoretical Framework and Mechanism

3.1 Environment and Agents

The economy has a single consumption good, a single period, and a competitive riskless rate of return R . There are N risk-neutral entrepreneurs, each endowed with an indivisible investment project requiring one unit of input and producing a random, non-negative, bounded output $\tilde{y}_i \in [0, \bar{y}]$. Each entrepreneur has zero personal wealth. Lenders (later, depositors) are also risk neutral, each endowed with wealth $1/m$, so financing one project requires pooling m lenders. The critical informational assumption is *ex-post* asymmetric information: the realization of $\tilde{y}$ is freely observed by the entrepreneur alone; any other agent can observe it only by paying a fixed monitoring cost $K > 0$. All parties agree that $E[\tilde{y}] > R + K$, so financing the project is efficient even net of monitoring costs, and $P(\tilde{y} = 0) > 0$, so the entrepreneur can never credibly promise a strictly positive payment with certainty.

3.2 The Direct Contract: Debt as the Optimal Response to Ex-Post Private Information

Because output is privately observed, any payment the entrepreneur makes must be self-enforcing: he will report whatever realization is best for him unless the contract punishes deviations. The paper introduces a *non-pecuniary* penalty function $\varphi(\cdot)$, interpreted as a bankruptcy cost the entrepreneur bears personally (time spent in proceedings, loss of reputation, and so on) that is not transferred to lenders – this is what allows an incentive-compatible contract to exist at all despite the entrepreneur's wealth constraint (he cannot pay more than he has). The optimal contract solves

$$\max_{\varphi(\cdot) \geq 0} E_{\tilde{y}} \int \max_{z \in [0, \tilde{y}]} \tilde{y} - z - \varphi(z) \quad (1a)$$

$$\text{subject to } z \in \arg \max_{z \in [0, \tilde{y}]} \tilde{y} - z - \varphi(z) \quad (1b)$$

$$\text{and } E[\arg \max_{z \in [0, \tilde{y}]} \tilde{y} - z - \varphi(z)] \geq R \quad (1c)$$

z	the payment the entrepreneur chooses to make to lenders, self-selected after observing $\tilde{y}$.
$\varphi(z)$	the non-pecuniary penalty triggered by choosing payment z ; it is a deadweight loss, not a transfer.
R	the competitive expected return lenders require, i.e. the market rate of return on the unit invested.

(1a) is the entrepreneur's ex-ante expected payoff maximized over penalty schedules; (1b) requires that, having observed $\tilde{y}$, the entrepreneur's own self-interested choice of z reproduces the contract's intended payment rule (incentive compatibility); (1c) is the lenders' zero-profit/competitive participation constraint. **PAPER** Proposition 1 solves this program in closed form: the optimal $\varphi^*(z) = \max(h - z, 0)$, where the face value h is the smallest solution to

$$P(\tilde{y} < h) \cdot E[\tilde{y} \mid \tilde{y} < h] + P(\tilde{y} \geq h) \cdot h = R \quad (2)$$

This is a standard debt contract: the entrepreneur pays $\min(\tilde{y}, h)$ and incurs a bankruptcy penalty exactly equal to the shortfall $h - \tilde{y}$ whenever $\tilde{y} < h$. The proof is by revealed-preference contradiction: any incentive-compatible penalty schedule that is not of this $\max(h - z, 0)$ form must impose a strictly larger penalty for some $z \leq h$ than the debt contract does, which is inconsistent with $\varphi(\cdot) \geq 0$ minimizing expected penalties subject to delivering R to lenders. **INFERENCE** Economically, (2) says the face value is set so that the lender's expected recovery – full recovery when solvent, partial (expected) recovery in the low tail when insolvent – exactly equals the required return; debt is optimal here for exactly the reason Townsend (1979) later formalizes generally: it is the contract that minimizes the region of the state space over which costly verification/penalty is triggered, because it only "activates" the penalty in the single lower tail where repayment capacity binds.

3.3 The Duplication Problem and the Case for Delegation

With costly monitoring available at cost K per observation, three institutional arrangements are possible: (i) no monitoring, the debt contract of Section 3.2 with expected deadweight cost $E[\varphi^*(\tilde{y})]$; (ii) each of the m lenders monitors independently, at aggregate cost mK ; (iii) monitoring is delegated to a single agent. Direct monitoring by all m lenders is worthwhile only if $mK \leq E[\varphi^*(\tilde{y})]$ – a condition that fails mechanically as m grows, because the loan is fixed in size (normalized to one) while the number of lenders needed to fund it, and hence the aggregate monitoring bill, grows with m . This is the paper's version of the free-rider logic: with dispersed small claims, no individual lender's share of the benefit of monitoring covers K , yet duplicating K across all of them is wasteful. Delegated monitoring is attractive whenever

$$K + D \leq \min[E[\varphi^*(\tilde{y})], m \cdot K] \quad (3)$$

where D is the *cost of delegation*: the expected deadweight loss required to give the monitor itself the incentive to monitor faithfully and to remit the proceeds honestly. The entire substantive question of the paper is reduced to characterizing D and showing when it is small.

3.4 Delegated Monitoring by a Financial Intermediary

The intermediary is itself a risk-neutral, zero-wealth agent. It raises deposits from mN depositors, lends to N entrepreneurs, monitors each at cost K , and receives unobserved payments $g_i(\tilde{y}_i)$ from each. Depositors observe only what the intermediary pays them, never the individual loan payments, the monitoring effort, or the entrepreneurs' true outcomes. This reproduces, one level up, exactly the incentive problem of Section 3.2: the intermediary can under-report its aggregate

receipts $\tilde{G}_N = \sum_{i=1}^N g_i(\tilde{y}_i)$ and keep the difference. Applying Proposition 1's logic to the intermediary-depositor relationship, the optimal deposit contract is again debt, with face value H_N solving the depositor-side analogue of (2), and expected delegation cost

$$D_N \text{ such that } e_{\tilde{G}_N}[\tilde{G}_N] - N \cdot K - H_N = \frac{1}{2} D_N \geq 0, \text{ with } H_N = N \cdot (R + D_N/2) \quad (4)$$

- $\tilde{G}_N$ aggregate payments the intermediary collects from its N borrowers, gross of monitoring costs.
- H_N face value of the intermediary's aggregate debt obligation to its depositors.
- D_N total expected delegation cost, i.e. the deadweight penalty the intermediary itself must be exposed to for its own incentive compatibility, per unit of N projects.

PAPER Proposition 2 is the central result of the paper: if the N project returns are independently distributed, $D_N \rightarrow 0$ as $N \rightarrow \infty$. The proof constructs the intermediary's payment schedule so that expected receipts equal $N(R + K + D_N)$, sets H_N at the midpoint implied by (4), and shows that the depositors' aggregate expected return exceeds $N \cdot R$ whenever the shortfall probability $P_N = P(\tilde{G}_N < H_N)$ is small enough, specifically for $P_N \in (0, (D_N/2)/(R + D_N/2))$. Because the individual project payoffs are bounded, i.i.d. (or merely independent), and each has expectation strictly above the amount owed, the weak law of large numbers drives the probability that the diversified pool's aggregate payment falls short of its promised debt service, P_N , to zero as $N \rightarrow \infty$. Since D_N is defined so as to make (4) hold with P_N as the driving term, $D_N \rightarrow 0$ follows. **INFERENCE** The mechanism is a pure law-of-large-numbers argument applied not to risk-sharing (everyone is risk neutral, so risk-sharing has no independent value here) but to the *probability of an incentive-relevant event* – the event that the pool cannot cover its promised debt service. Diversification does not make the intermediary safer in a utility sense; it makes default (and hence the occasions on which the intermediary must be "caught" understating and penalized) vanishingly rare, which is precisely what a moral-hazard-based delegation cost needs to shrink.

3.5 Correlated Returns and the Link to the Arbitrage Pricing Theory

Independence across all N borrowers is a strong assumption. The paper's Corollary to Proposition 2 relaxes it to a factor structure,

$$\tilde{y}_i = \sum_{j=1}^M \beta_{ij} F_j + \varepsilon_i \quad (5)$$

where the factors F_j are commonly observable ex post (e.g. GNP, interest rates, input prices) and the idiosyncratic terms ε_i are independent across borrowers. **PAPER** Diamond shows $D_N \rightarrow 0$ still holds provided the intermediary hedges its exposure to the F_j – either via futures markets or via deposit contracts contingent on the observable factors – and retains only the diversifiable idiosyncratic risk. **EXTENSION** The paper notes explicitly that this return-generating process "is consistent with the Arbitrage Pricing Theory of Ross (1976)"; the deeper point, not fully spelled out in the paper, is that Proposition 2 requires diversifiability of the *specific* component of loan returns, exactly the decomposition APT relies on to price only factor risk. This gives a rationale, independent of portfolio-CAPM risk-pricing logic, for why banks hedge interest-rate and macro exposures (matched maturities, swaps, futures) while retaining idiosyncratic credit risk on

balance sheet: the idiosyncratic component is what diversification can neutralize for incentive purposes, while the systematic component cannot be diversified away and must instead be shared or hedged explicitly.

3.6 Risk Aversion and Two Kinds of Diversification

Section 4 of the paper drops the wealth constraint (so debt/bankruptcy penalties are no longer mechanically necessary) and asks whether diversification still helps when agents are risk averse in the ordinary Arrow-Pratt sense. Two conceptually distinct forms of diversification are compared.

DIVERSIFICATION BY SUBDIVIDING RISK

If N risk-averse bankers each retain a $1/N$ share of every loan (and can observe each other's monitoring, so there is no within-intermediary incentive problem), each banker's risk exposure shrinks with N – this is ordinary portfolio diversification/risk-sharing, and it unambiguously reduces each banker's risk premium as $N \rightarrow \infty$ (Ramakrishnan and Thakor (1984)).

DIVERSIFICATION BY ADDING RISK

If instead a *single* banker takes on more independent loans as N grows – bearing 100% of each, with no subdivision – total risk borne by that one agent is increasing in N , not decreasing. Samuelson (1963) called the presumption that this still helps a "fallacy of large numbers." Diamond derives when it is not, in fact, a fallacy. For two independent bounded gambles $\tilde{x}_1, \tilde{x}_2$ with risk premia π_1, π_2 defined by $E[U(W_0 + \tilde{x}_i + \pi_i)] = U(W_0 + E[\tilde{x}_i])$, and joint premium π_{1+2} defined analogously for $\tilde{x}_1 + \tilde{x}_2$, adding risks helps – $\pi_{1+2} < \pi_1 + \pi_2$ – if and only if the "indirect" utility function obtained by folding $\tilde{x}_1$ into wealth is less risk averse than $U(\cdot)$ itself. A sufficient condition, obtained via Jensen's inequality under $E[\tilde{x}_1] \approx 0$, is

$$U''''(W) \geq 0 \text{ and } U''''''(W) \leq 0 \text{ (one strict), jointly with decreasing absolute risk aversion: } U''(W) \cdot U''''''(W) > U''''(W)^2 \quad (6)$$

PAPER Constant absolute risk aversion (exponential utility) and quadratic utility both fail this condition – for CARA agents, adding independent gambles never helps or hurts, because risk aversion toward the marginal gamble does not depend on wealth already at risk; for quadratic utility (increasing absolute risk aversion once a positive-mean gamble is voluntarily accepted), adding risks strictly hurts. **CRITIQUE** This is an important qualifier the paper is candid about: unlike the risk-neutral result of Section 3, which needs no assumption on preferences beyond risk neutrality and a wealth constraint, the risk-averse "diversification by adding risk" result is fragile to functional form and, as Diamond notes explicitly, essentially untestable given that preferences are unobservable. The paper's own conclusion (Section 4, final paragraph) is that the risk-neutral wealth-constrained result of Section 3 is the "strong and testable" one; the risk-averse extension is best read as establishing that the risk-neutral result is not an artifact of risk neutrality per se, not as an independent, equally robust channel.

3.7 Reconciliation with Leland-Pyle (1977)

Leland and Pyle model *ex-ante* adverse selection: the entrepreneur privately knows the mean quality of the project before financing, and signals quality by retaining equity (bearing costly idiosyncratic risk under CARA utility and normal returns, valued via a CAPM-consistent market). Diamond imports diversification into this setting with N independent projects and shows the same subdivide-versus-add distinction applies: an intermediary that adds together several projects under a single CARA decision-maker gets no signalling-cost reduction (Section 3.6's negative case exactly, since CARA is preference-invariant to independent additions), while an intermediary of N bankers who each retain $1/N$ of every project's equity *does* lower the per-project signalling cost, because each banker's problem becomes equivalent to signalling a single project with variance shrunk by $1/N$. In deriving this, Diamond finds and corrects a sign error in Leland-Pyle's Proposition III: their formula omits a term (an a^2 component) whose absence made the entrepreneur's expected utility appear to be increasing, rather than decreasing, in project variance – a result the corrected derivative,

$$dE[U(W_p)]/d\sigma_x^2 = -b[(1-a)(\log(1-a)+a) + a^2/2] < 0 \text{ for } a \in (0,1) \quad (7)$$

overturns. **PAPER** With the corrected sign, diversification by subdividing risk unambiguously lowers signalling costs in the Leland-Pyle setting as well, so the two very different models of intermediation (costly-monitoring/moral hazard here, and costly-signalling/adverse selection in Leland-Pyle) deliver the same qualitative conclusion: intermediation is viable when diversification is organized as risk-subdivision among informed insiders, not as simple risk-pooling by a single decision-maker.

04. Model Predictions and Results

Because the paper is purely theoretical, its "results" are testable predictions and comparative statics rather than estimated coefficients. Four predictions follow directly from Propositions 1 and 2 and are treated in the text as the model's connection to observed banking practice.

HIGH LEVERAGE, LOW DEFAULT PROBABILITY

The intermediary's optimal capital structure is overwhelmingly debt (deposits), with face value $H_N = N \cdot (R + D_N/2)$ close to the expected future value of its assets $N \cdot (R + K + D_N)$ as N grows, yet the probability of default, P_N , is driven to zero by the same limit that shrinks D_N . **INFERENCE** This is a distinctive and non-obvious prediction: high leverage and low default risk are not in tension in this model, because leverage here is not a risk-shifting device but the optimal incentive instrument, and diversification is what keeps its downside from materializing. This matches the observed fact that commercial banks operate with far higher leverage than non-financial firms while historically exhibiting relatively low steady-state default rates outside systemic episodes.

DEBT, NOT EQUITY, AT BOTH LAYERS

Both the borrower-to-intermediary loan contract and the intermediary-to-depositor deposit contract are debt with non-pecuniary bankruptcy penalties. **PAPER** The paper explicitly states this is a consequence of applying Proposition 1's logic twice, once at each contracting layer, and is why bank loans and bank deposits both take debt form in the model even though nothing in the environment forces a specific institutional label onto either instrument.

ILLIQUID BANK LOANS

PAPER Because the value of the intermediary's information about a specific loan is private and non-transferable without re-incurring the monitoring cost K , loans monitored by one intermediary are not easily sold to another (adverse selection in which loans get sold would also arise). **INFERENCE** This gives a demand-side-independent, information-based account of loan illiquidity that complements Diamond-Dybvig's supply-side account (illiquidity built into the technology); read together, the two papers argue that bank loans are illiquid both because monitoring information cannot be costlessly transferred (this paper) and because banks intentionally hold long-gestation projects funded by short liabilities (Diamond-Dybvig).

A RATIONALE FOR HEDGING SYSTEMATIC RISK

Following directly from the Corollary in Section 3.5, the model predicts intermediaries should hedge or contractually pass through observable systematic risk (interest rate risk, macro factors) while retaining idiosyncratic credit risk for monitoring purposes. **PAPER** The paper frames this as a direct contribution to the then-live regulatory debate over whether banks should be permitted to use interest-rate futures, arguing the model supports permitting such hedging.

Table 1. Summary of the model's central comparative-static result

Quantity	Behaviour as $N \rightarrow \infty$	Economic interpretation
Delegation cost per project, D_N	$\rightarrow 0$	Cost of providing the intermediary incentives to remit honestly vanishes
Shortfall probability, $P_N = P(\tilde{G}_N < H_N)$	$\rightarrow 0$	Weak law of large numbers on independent (or factor-purged) project returns
Face value of deposit debt, H_N	$\rightarrow N \cdot (R + K)$	Depositor claim converges to the riskless value of net monitored proceeds
Leverage of the intermediary	high, and rising	Optimal incentive instrument, not a risk-shifting distortion

Source: constructed by the reviewer from Propositions 2 and its proof (Diamond, 1984, Section 3).

05. Robustness and Extensions Within the Paper

As a theoretical paper, "robustness" here means the set of alternative assumptions the author checks the central result against, rather than alternative empirical specifications. Table 2 organizes these using a concern → test → result → adequacy structure.

Table 2. Robustness checks performed within the paper

Concern	Test	Result	Does it resolve the concern?
Independence of project returns is unrealistic	Factor-structure corollary (Section 3.5, eq. 5)	$D_N \rightarrow 0$ survives if systematic factors are hedged and only the idiosyncratic residual is diversified	Partially – requires that systematic factors be both observable and hedgeable; says nothing about diversification if factor exposures cannot be hedged
Result depends on risk neutrality	Risk-averse extension (Section 4)	Diversification still helps under "subdividing"; helps under "adding" only for a restrictive utility class	Yes for subdividing risk; No in general for adding risk – the qualifier is disclosed rather than concealed
Result might be an artifact of the specific monitoring/moral-hazard model	Cross-check against Leland-Pyle's adverse-selection/signalling model (Section 3.7)	Same qualitative conclusion (subdividing diversification lowers costs) obtains in a structurally different model, once L-P's own algebra is corrected	Yes – strengthens confidence that the result is about diversification generically, not an artifact of the monitoring formalism
Single intermediary with one borrower	$N = 1$ boundary case (Section 3.4)	Shown formally non-viable – delegation cost at least as large as direct monitoring, plus the physical cost K on top	Yes, and usefully falsifiable: the model does not merely assert diversification helps, it shows the boundary case where intermediation is strictly dominated

Source: constructed by the reviewer from Diamond (1984), Sections 3–5 and Appendix.

CRITIQUE What the paper does not stress-test is arguably more informative than what it does. There is no analysis of what happens if depositors can observe a noisy public signal of the intermediary's performance (which would blend this model with reputation-based theories), no analysis of what happens if the intermediary itself needs to be recapitalized after a bad realization (a dynamic, multi-period question the one-period model cannot address), and no analysis of intermediary-level runs, which the companion Diamond-Dybvig (1983) paper takes up separately with a different technology (sequential service, liquidity shocks) rather than as an extension of this model's own incentive structure.

06. Critical Evaluation

6.1 What the Paper Establishes

ESTABLISHED

Under ex-post asymmetric information, wealth-constrained risk-neutral entrepreneurs, and a fixed physical monitoring cost K , debt with a non-pecuniary bankruptcy penalty is the cost-minimizing contract between a single principal and a single wealth-constrained agent (Proposition 1). This is a clean, general, and subsequently much-reproduced result (it is the deterministic special case of Townsend's (1979) and Gale-Hellwig's (1985) costly state verification results).

ESTABLISHED, CONDITIONAL ON INDEPENDENCE OR HEDGEABLE FACTOR STRUCTURE

The per-borrower cost of delegating monitoring to a diversified intermediary vanishes as the number of independent (or factor-purged) borrowers grows (Proposition 2). This is a rigorous limiting result, proved from first principles using the weak law of large numbers, and it is the paper's central and most durable contribution.

SUPPORTED BUT CONDITIONAL

Diversification also helps under risk aversion, but only for the "subdividing" mode of diversification and, for the "adding" mode, only under a specific and empirically unverifiable class of utility functions (decreasing absolute risk aversion with $U'''' > 0$, $U'''''' \leq 0$). The paper is transparent that this channel is not testable in the way the risk-neutral channel is.

NOT ESTABLISHED

The paper does not establish that intermediation is *optimal* in any mechanism-design sense (i.e. that debt-financed diversified monitoring is the best of *all* conceivable institutional arrangements, rather than the best among the specific set the author compares). It also does not establish anything about the intermediary's own stability, run risk, or capital adequacy over time; those questions are explicitly outside the model's one-period, single-intermediary scope.

6.2 Strengths

CRITIQUE The paper's greatest strength, recognized by the author himself in the introduction, is that it closes the analytical gap left by prior intermediation theories: it is the first paper to net the cost of controlling the intermediary's own incentive problem against the gross benefit of delegated monitoring, rather than assuming away the intermediary's own agency problem. Independently of the author's framing, three further strengths stand out. First, the central result requires remarkably weak assumptions – risk neutrality, a wealth constraint, and independence (or a

hedgeable factor structure) – and delivers a sharp, closed-form limiting result, which is unusual economy in a theory paper. Second, the model is genuinely falsifiable in its comparative statics: it predicts specific joint behaviour of leverage and default probability that differs from risk-shifting or tax-shield theories of bank capital structure. Third, the paper explicitly derives, rather than assumes, the debt form of both the loan and the deposit contract from the same underlying primitive (costly ex-post verification), giving a unified microfoundation for two contracts that competing theories usually take as given.

6.3 Weaknesses and Critique

CRITIQUE Several independent weaknesses merit attention beyond what the paper itself flags.

Single-intermediary, closed-economy assumption. The model has one intermediary per pool of borrowers and does not ask what happens when multiple competing intermediaries can bid for the same borrowers, or when a borrower can be monitored by more than one intermediary. This matters because competition among intermediaries could, in principle, erode the very informational rents that make delegated monitoring incentive-compatible in the first place; the paper is silent on whether Proposition 2's cost advantage survives entry.

Static, one-period horizon. Because there is only one period, the model cannot speak to how an intermediary rebuilds capital after a bad draw, how repeated interaction (reputation) might substitute for or reinforce diversification, or how the delegation-cost logic changes once depositors can withdraw early – precisely the question Diamond-Dybvig (1983) takes up, but with a different (liquidity-shock) technology rather than as a dynamic extension of this model.

No monitoring of the monitor's monitoring quality. The model assumes the intermediary, once given the correct incentive to remit payments truthfully, also monitors truthfully and does not shirk on monitoring effort itself; Diamond shows the same contract that induces truthful remittance happens to also induce optimal monitoring effort (Section 3.4), but this "two birds, one contract" result relies on the specific functional form chosen and is not obviously robust to richer effort-choice environments (e.g. monitoring intensity chosen along a continuum with a separate cost function).

Diversification benefit is entirely idiosyncratic-risk-based. The Corollary of Section 3.5 requires that systematic risk be both observable and hedgeable. Where these conditions fail – opaque, correlated, or non-hedgeable systematic exposures, of exactly the kind that generated the 2008 financial crisis in mortgage-backed pools – the model's central mechanism does not merely weaken, it can reverse: correlated tail risk in a "diversified" portfolio can make P_N fail to shrink with N , or shrink far more slowly than the i.i.d. benchmark implies, precisely because default clustering breaks the law-of-large-numbers logic Proposition 2 relies on.

The single most important weakness is this last point: the model's central prediction – low default probability despite high leverage – is derived under an assumption (return independence, or at minimum hedgeable factor structure) that is a testable but very strong empirical claim about loan portfolios, and one that is least likely to hold precisely during the systemic episodes when the prediction matters most for financial stability. The paper is honest that the result is conditional on

independence, but does not explore how fast delegation costs deteriorate as correlation rises from zero, leaving the practically most important comparative static – "how much correlation before intermediation stops paying?" – unanswered within the paper itself.

07. Contradictions and Edge Cases

EXTENSION Pushing the model to its boundaries clarifies exactly where the mechanism can fail.

N = 1: INTERMEDIATION IS STRICTLY DOMINATED

The paper itself proves this case: with a single borrower, the intermediary's delegation cost is at least as large as the direct deadweight cost of contracting without an intermediary, plus the intermediary now also spends K on monitoring. Diversification is not merely helpful at the margin here, it is *necessary* for intermediation to be viable at all – a genuinely falsifiable prediction: single-borrower "intermediaries" (a lender who lends to exactly one firm and nobody else) should never be observed as viable institutions in this framework, which is broadly consistent with the empirical non-existence of single-borrower banks.

PERFECT CORRELATION: THE MECHANISM SWITCHES OFF ENTIRELY

If all N project returns are perfectly correlated, pooling N of them is informationally equivalent to holding one large project scaled by N ; the shortfall probability P_N does not shrink with N at all, so D_N does not converge to zero and intermediation confers no advantage over direct monitoring at the margin. **INFERENCE** This edge case is the cleanest illustration that the model's engine is diversification of idiosyncratic risk specifically, not "size" or "scale" per se – a large intermediary lending to many correlated borrowers gets none of Proposition 2's benefit, which is a useful corrective to a common informal argument that "bigger banks are safer because they are more diversified."

K → 0: DELEGATION BECOMES MOOT

As the physical monitoring cost K approaches zero, the value of any monitoring arrangement (direct or delegated) collapses toward zero as well, since the deadweight bankruptcy cost under the no-monitoring contract already shrinks to nothing when verification is nearly free. In this limit the entire delegation-cost apparatus becomes economically irrelevant, which is a useful sanity check: the model correctly predicts that intermediation should matter most exactly where information is hardest (most costly) to verify, i.e. for opaque, hard-to-value borrowers such as small and mid-sized firms, consistent with the empirical fact that bank-dependent borrowers tend to be smaller, younger, and less transparent than bond-market borrowers.

EXTREME LEVERAGE, H_N NEAR THE ASSET'S EXPECTED VALUE

As N grows, H_N approaches $N \cdot (R+K)$, i.e. the intermediary's promised debt service converges to (nearly) the full expected value of its monitored assets. **CRITIQUE** Taken to its logical extreme this predicts a bank should hold negligible equity capital, which is empirically false (banks hold meaningfully more capital than the model's asymptotic limit) and points to the paper's silence on

what, outside the model, pins down a positive capital buffer – most plausibly the very run risk and liquidity considerations analyzed in the companion Diamond-Dybvig (1983) paper, or ex-ante regulatory capital requirements the model does not include.

COULD ANOTHER MECHANISM GENERATE THE SAME PREDICTIONS?

CRITIQUE High leverage combined with (in normal times) low realized default rates is also consistent with deposit-insurance-induced risk shifting (Merton (1977)) or with regulatory capital arbitrage, both of which predict high leverage for reasons entirely unrelated to delegated monitoring. The paper does not offer, and by its static one-period design cannot offer, a way to empirically distinguish "high leverage because diversification makes debt incentive-efficient" from "high leverage because deposit insurance subsidizes it," which is a live identification problem for anyone trying to test Proposition 2's predictions against real bank balance sheets.

08. Further Literature

FOUNDATIONS THE PAPER BUILDS ON

Townsend (1979), "Optimal Contracts and Competitive Markets with Costly State Verification" → shared idea: privately observed output, costly verification → relationship: Townsend's general costly-state-verification framework (with stochastic verification) nests Diamond's Proposition 1 as the deterministic-verification special case → why it matters: it situates Diamond's debt-optimality result within the broader CSV literature and clarifies which of Diamond's assumptions (deterministic, not stochastic, verification) are restrictive.

Harris and Raviv (1979); Holmstrom (1979); Shavell (1979) → shared idea: single-principal, single-agent monitoring resolves moral hazard → relationship: these supply the one-lender benchmark the paper extends to many lenders and a delegated monitor → why it matters: without them, the paper's claim that "delegation creates its own incentive problem" would lack a clean baseline to net against.

Leland and Pyle (1977), "Informational Asymmetries, Financial Structure, and Financial Intermediation" → shared idea: costly signalling can motivate financial intermediation → relationship: Diamond directly re-derives and corrects L-P's Proposition III, showing the same subdivide-diversification result holds in their adverse-selection framework → why it matters: this cross-model agreement is the strongest evidence in the paper that the diversification result is not an artifact of the specific moral-hazard formalism.

CONTEMPORANEOUS COMPANION

Diamond and Dybvig (1983), "Bank Runs, Deposit Insurance, and Liquidity" (already reviewed in this library) → shared idea: banks as institutions holding illiquid assets funded by liquid liabilities → relationship: complementary, not overlapping – this paper explains why intermediation is valuable (diversification lowers delegation costs); Diamond-Dybvig explains why the resulting

institution is fragile (sequential-service demand deposits admit a self-fulfilling run equilibrium) → why it matters: read together they give both the benefit and the cost side of the bank balance-sheet structure that neither paper alone addresses.

PAPERS THAT DEVELOPED THE CONTRIBUTION FURTHER

Ramakrishnan and Thakor (1984), “Information Reliability and a Theory of Financial Intermediation” → shared idea: diversification within an intermediary of risk-averse agents → relationship: formalizes the "subdividing risk, no internal incentive problem" case Diamond sketches informally in Section 4 → why it matters: extends the risk-averse robustness check the present paper only partially develops.

Boyd and Prescott (1986), “Financial Intermediary-Coalitions” → shared idea: coalitions of agents can support information-based intermediation without a single unmonitored delegate → relationship: offers an alternative micro-mechanism (coalition formation) to Diamond's single-delegate structure, addressing the "who monitors the monitor" gap identified in Section 6.3 above → why it matters: shows Diamond's specific institutional form (one intermediary, delegated fully) is not the only way to reach a similar diversification benefit.

Diamond (1991), “Monitoring and Reputation: The Choice Between Bank Loans and Directly Placed Debt” → shared idea: intermediated monitoring versus direct finance → relationship: adds a reputation dimension and a borrower life-cycle (bank-dependent early, direct-market-access later) largely absent from the static 1984 model → why it matters: dynamizes the choice between intermediated and direct finance that the 1984 paper treats only as a one-shot comparison.

Holmström and Tirole (1997), “Financial Intermediation, Loanable Funds, and the Real Sector” → shared idea: intermediary capital and monitoring incentives → relationship: endogenizes the intermediary's own capital structure and shows monitoring incentives depend on the intermediary's own net worth, addressing the "what pins down a positive capital buffer" gap noted in Section 7 → why it matters: supplies the capital-structure determinant the 1984 model's asymptotic $H_N \rightarrow N(R+K)$ limit leaves unexplained.

Winton (1995), “Delegated Monitoring and Bank Structure in a Finite Economy” → shared idea: delegated monitoring with a finite number of borrowers → relationship: relaxes the $N \rightarrow \infty$ asymptotic and studies the finite- N delegation cost directly, including cases with correlated returns → why it matters: directly answers the "how fast does D_N deteriorate with correlation" question flagged in Section 6.3 as unaddressed by the original paper.

Krasa and Villamil (1992), “Monitoring the Monitor: An Incentive Structure for a Financial Intermediary” → shared idea: revisits whether depositors need to monitor the intermediary → relationship: directly engages the "who monitors the monitor" weakness identified in Section 6.3, showing conditions under which the delegated-monitoring result is robust to depositor monitoring costs → why it matters: closes a gap the 1984 paper leaves open by fiat (assuming depositors simply cannot observe the intermediary).

09. Research Gaps

9.1 Paper-Level Gaps

The comparative static of delegation cost with respect to correlation is not derived; only the two polar cases (independence, perfect correlation) are characterized. The model is silent on intermediary entry/competition, on multi-period capital dynamics, and on what determines the equilibrium number and size distribution of intermediaries.

9.2 Literature-Level Gaps

Even across the literature that followed (Boyd-Prescott, Holmström-Tirole, Winton), there is no fully unified model that simultaneously endogenizes (i) the intermediary's optimal size and diversification, (ii) its capital structure and buffer against correlated shocks, and (iii) its exposure to Diamond-Dybvig-style run risk, within one coherent dynamic framework. Post-2008 systemic-risk models (e.g. network models of interbank contagion) address correlation and contagion but rarely reconnect to the underlying delegated-monitoring microfoundation this paper establishes.

9.3 Methodological Gaps

Gap → no empirical estimate of how quickly real loan-portfolio delegation costs fall with diversification. **Why it matters** → Proposition 2 is an asymptotic result; practitioners need the finite-N rate of convergence to judge how many independent borrowers are "enough." **Existing limitation** → loan-level default correlation is hard to observe directly and is typically inferred from reduced-form credit-risk models (e.g. CreditMetrics-style default correlation), not derived from a delegated-monitoring structural model. **Possible way forward** → structural estimation of a Diamond-style delegation-cost function using granular small-business loan data with plausibly exogenous variation in a bank's borrower-pool diversification (e.g. geographic or sectoral).

10. Research Opportunities

Idea 1 — Finite-N delegation costs and bank size in small-business lending data

RESEARCH QUESTION How quickly does the empirical wedge between a bank's cost of funds and its lending rate (a proxy for delegation cost) fall as a bank's small-business borrower pool grows and diversifies across sectors and geographies?

MOTIVATION Proposition 2 is asymptotic; the finite-N convergence rate is untested and directly policy-relevant for assessing minimum viable community-bank size.

HYPOTHESIS The delegation-cost proxy falls in borrower-pool size at a rate consistent with a law-of-large-numbers curve, flattening once sectoral/geographic correlation dominates idiosyncratic variance.

ECONOMIC MECHANISM Diamond's Proposition 2, extended to finite N with a factor structure as in the Corollary.

LITERATURE CONNECTION Extends Winton (1995); complements bank-lending-relationship literature (Petersen and Rajan).

DATA Call Report data (U.S. FFIEC) matched to CRA small-business lending disclosures, or comparable European AnaCredit loan-level data.

VARIABLES Bank-level loan spread net of funding cost; Herfindahl index of the small-business loan book by sector and county; bank size and capital ratio controls.

METHOD Panel regression with bank and time fixed effects; instrument diversification with merger-induced geographic footprint changes.

IDENTIFICATION STRATEGY Exploit bank mergers that mechanically expand borrower-pool diversification for reasons plausibly unrelated to a given bank's own lending-spread decisions.

EXPECTED CONTRIBUTION First direct empirical mapping from Diamond's delegation-cost construct to observed loan pricing.

MAIN RISK Loan spreads reflect many frictions beyond delegation cost (market power, risk-based pricing), making the mapping noisy.

Idea 2 — Correlation-adjusted delegation cost during systemic stress

RESEARCH QUESTION How much of the funding-cost spike faced by highly diversified banks during 2008 and 2020 is explained by a spike in realized cross-borrower correlation, as Diamond's model would predict (Section 7 above)?

MOTIVATION Directly tests the edge case identified in Section 7: the model's diversification benefit switches off when correlation rises, exactly when systemic risk premia should rise.

HYPOTHESIS Banks whose pre-crisis borrower pools display higher realized default correlation during the crisis window experience larger increases in wholesale funding spreads, controlling for capital and size.

ECONOMIC MECHANISM Rising default correlation raises P_N (shortfall probability) even for large N , raising the intermediary's own delegation/default risk from the depositor's perspective.

LITERATURE CONNECTION Connects Diamond (1984) to the interbank contagion and systemic risk literature (Allen and Gale (2000); Acharya (2009)).

DATA U.S. bank holding company call reports, syndicated loan (DealScan) default correlations, CDS spreads on bank holding companies 2006–2010 and 2019–2021.

VARIABLES Realized default correlation within a bank's loan book; bank CDS spread; leverage; loan-book Herfindahl.

METHOD Event-study/panel regression around the two crisis windows.

IDENTIFICATION STRATEGY Pre-crisis correlation measured on a pre-determined loan book, avoiding reverse causality from crisis-period funding stress to portfolio composition.

EXPECTED CONTRIBUTION Direct empirical test of the model's own stated boundary condition, rather than of its untested asymptotic case.

MAIN RISK Default correlation is itself endogenous to the crisis, complicating a clean pre-determined measure.

Idea 3 — A dynamic synthesis of Diamond (1984) and Diamond-Dybvig (1983)

RESEARCH QUESTION In a multi-period model combining delegated-monitoring incentive costs with sequential-service liquidity demand, what capital buffer does the intermediary hold in equilibrium, and how does it compare to the near-zero buffer implied by the 1984 paper's asymptotic limit?

MOTIVATION Section 7 above shows the 1984 model's asymptotic H_N predicts implausibly low capital; the companion 1983 paper's run risk is a natural candidate to pin down a positive buffer, but the two models have never been formally merged.

HYPOTHESIS Optimal capital buffer is increasing in both the correlation of borrower returns and the probability of a Diamond-Dybvig-style panic-run equilibrium.

ECONOMIC MECHANISM Capital serves jointly as an incentive device (delegation cost) and a liquidity/run buffer (Diamond-Dybvig), two roles the separate 1983/1984 papers do not jointly optimize.

LITERATURE CONNECTION Builds on Holmström and Tirole (1997) and the bank-capital-regulation literature (Admati and Hellwig).

DATA Primarily a calibration/computational exercise; disciplined with FDIC failed-bank capital ratios as a target moment.

VARIABLES Model-implied optimal capital ratio versus observed pre-failure capital ratios of failed community banks.

METHOD Dynamic contract-theoretic model solved numerically; calibration/simulation, not reduced-form estimation.

IDENTIFICATION STRATEGY Not applicable in the causal-inference sense; internal validity rests on matching calibration targets.

EXPECTED CONTRIBUTION A genuinely unified microfoundation for bank capital that neither 1983 nor 1984 paper alone provides.

MAIN RISK High modelling complexity; results may be sensitive to functional-form choices at the two-model interface.

Idea 4 — Fintech marketplace lenders as a test of the delegation-cost theory without balance-sheet intermediation

RESEARCH QUESTION Do marketplace/peer-to-peer lenders, which diversify investors directly across many independently-originated loans without the platform itself holding the credit risk, achieve the same asymptotic reduction in effective borrowing cost that Diamond's balance-sheet intermediary achieves – or does removing the platform's own "skin in the game" reintroduce a first-layer delegation cost the model predicts should not vanish?

MOTIVATION Marketplace lending is a natural experiment in whether Diamond's diversification result requires the intermediary itself to bear default risk (as in the model) or can be replicated by pass-through diversification to end investors.

HYPOTHESIS Marketplace platforms that retain a first-loss tranche (bearing some delegation cost, closer to Diamond's structure) exhibit lower investor-required yields, for a given borrower pool quality, than platforms with pure pass-through and no retained risk.

ECONOMIC MECHANISM Retained first-loss exposure recreates the intermediary's own incentive-compatibility constraint that Proposition 2 shows diversification can satisfy at low cost; pure pass-through platforms instead push the un-diversified verification problem back onto atomized investors.

LITERATURE CONNECTION New empirical application of Diamond (1984) to fintech, connecting to the recent literature on marketplace lending risk retention (Balyuk, Vallee).

DATA LendingClub/Prosper loan-level performance and platform-structure data; European P2P platforms with varying risk-retention rules.

VARIABLES Platform risk-retention share; investor required yield; realized default correlation within platform's loan pool.

METHOD Cross-platform panel regression, exploiting regulatory differences in mandated risk retention (e.g. EU risk-retention rules versus unregulated U.S. platforms) as a source of variation.

IDENTIFICATION STRATEGY Regulatory risk-retention mandates as a quasi-experimental source of variation in retained exposure, orthogonal to individual platform credit quality.

EXPECTED CONTRIBUTION Tests whether Diamond's mechanism is about diversification per se or about diversification *combined with* retained, incentive-relevant exposure – a distinction the 1984 model does not itself separate.

MAIN RISK Platform choice of risk retention is endogenous to unobserved loan quality, confounding the comparison.

Idea 5 — Extending Proposition 2 to a network of correlated intermediaries (systemic delegation cost)

RESEARCH QUESTION When many intermediaries each individually satisfy Diamond's diversification condition but their loan pools are cross-correlated through shared macro exposure or common counterparties, does the system-wide probability of joint shortfall still vanish, or does systemic correlation impose an aggregate delegation cost no individual intermediary internalizes?

MOTIVATION Directly generalizes the perfect-correlation edge case in Section 7 to a multi-intermediary, network setting, connecting the paper to macroprudential regulation.

HYPOTHESIS System-wide shortfall probability across intermediaries fails to converge to zero even as each individual intermediary's own $D_N \rightarrow 0$, once common factor exposure across intermediaries is introduced.

ECONOMIC MECHANISM Diamond's Corollary (Section 3.5) already shows within-intermediary diversification requires hedgeable idiosyncratic risk; extending to across-intermediary correlation shows an externality individual intermediaries do not price.

LITERATURE CONNECTION Bridges Diamond (1984) to the systemic risk and macroprudential capital literature (Acharya, Pedersen, Philippon, Richardson (2017)).

DATA Primarily theoretical; illustrative calibration with BIS cross-border banking claims data for a systemic factor structure.

VARIABLES N/A (theoretical); calibration targets systemic factor loadings from cross-bank stock/CDS return correlations.

METHOD Extend the Corollary's factor model (eq. 5) to a multi-intermediary system; derive conditions for a macroprudential capital surcharge that internalizes the externality.

IDENTIFICATION STRATEGY Not applicable (theoretical); empirical companion piece would use the same cross-bank correlation approach as Idea 2.

EXPECTED CONTRIBUTION A microfounded rationale, derived from Diamond's own delegation-cost logic, for systemic capital surcharges on interconnected/correlated intermediaries.

MAIN RISK Purely theoretical extension risks being seen as incremental unless paired with a calibration exercise with genuine empirical bite.

11. Research Idea Ranking

Ranked by a combination of novelty, feasibility, data availability, methodological quality, economic importance, and expected contribution:

TOP THREE

1. Idea 2 (Correlation-adjusted delegation cost during systemic stress). Highest ranked because it directly tests the paper's own explicitly acknowledged boundary condition using readily available crisis-period data (CDS spreads, call reports), has clean identification via pre-determined loan-book correlation, and speaks to a first-order policy question (systemic risk pricing).

2. Idea 1 (Finite-N delegation costs in small-business lending). Strong because it operationalizes the paper's central asymptotic result with the most directly available data (Call Reports, CRA disclosures) and a plausible merger-based instrument, though the mapping from loan spreads to "delegation cost" is noisier than Idea 2's more crisis-focused design.

3. Idea 4 (Fintech marketplace lenders). Novel and timely – marketplace lending offers a rare natural laboratory to separate diversification from retained incentive-relevant exposure, a distinction the original paper does not itself test – but is somewhat more exposed to endogenous-platform-selection concerns than Ideas 1 and 2.

Ideas 3 and 5 are ranked lower for this stage not because they are less important – if anything Idea 3's synthesis of Diamond (1984) and Diamond-Dybvig (1983) addresses the single biggest unresolved tension identified in this review (near-zero implied capital versus observed capital buffers) – but because both are primarily theoretical/computational projects with longer horizons and no immediate empirical test, making them better suited as a second-stage extension once the empirical groundwork from Ideas 1 and 2 is in place.

12. Learning Framework

Prerequisite Concepts

Basic principal-agent theory and moral hazard (Holmstrom (1979)); the concept of a non-pecuniary/deadweight penalty as distinct from a monetary transfer; the weak law of large numbers and its use in incentive (not just risk-sharing) arguments; Arrow-Pratt risk aversion and the certainty-equivalent/risk-premium apparatus; basic mean-variance/CAPM intuition (used in the Leland-Pyle comparison of Section 3.7).

Core Concepts Learned

The distinction between the *gross* benefit of delegating information production and the *net* benefit after accounting for the delegate's own incentive problem (the "delegation cost"); the distinction between diversification as risk-sharing (relevant under risk aversion) and diversification as probability-of-default reduction (relevant even under risk neutrality, given a wealth constraint); the distinction between "subdividing" and "adding" risk (Samuelson's fallacy of large numbers) and why only the former is robust to preference specification.

Important Equations

The debt-contract characterization (2); the delegation-cost viability condition (3); the asymptotic delegation-cost result underlying Proposition 2 (4); the factor-structure Corollary (5); the sufficient condition for beneficial risk-adding diversification (6).

Methods Learned

Mechanism design under a wealth constraint (as opposed to unconstrained moral hazard); asymptotic (limit- N) argumentation using the weak law of large numbers as an incentive-cost-reduction device rather than a pure risk-diversification device; comparative use of Jensen's inequality to sign a risk-premium comparison; cross-model robustness checking by re-deriving a result inside a structurally different model (moral hazard versus adverse selection/signalling).

Questions to Test Understanding

Comprehension – Why does direct monitoring by all m lenders become infeasible as m grows, even though the aggregate benefit of monitoring, $E[\varphi^*(\tilde{y})]$, does not depend on m ? **Analysis** – Why does Proposition 2's result depend on the weak law of large numbers acting on the probability of a shortfall event, rather than on any reduction in the variance of the intermediary's own utility (which would require risk aversion)? **Research** – Under what precise conditions on the correlation structure of the factor loadings β_{ij} in equation (5) does the Corollary's $D_N \rightarrow 0$ result survive, and can you derive the rate of convergence as a function of the largest factor loading?

Papers to Read Next

Townsend (1979), “Optimal Contracts and Competitive Markets with Costly State Verification”, for the general costly-state-verification framework this paper's Proposition 1 specializes; Diamond and Dybvig (1983), already in this library, for the companion liquidity/run-risk side of the bank balance sheet; Holmström and Tirole (1997), “Financial Intermediation, Loanable Funds, and the Real Sector”, for the dynamic capital-structure question this paper leaves open.

13. Conclusion

ONE-SENTENCE THESIS

Financial intermediaries exist because diversification across many independently monitored borrowers drives the cost of giving the intermediary itself incentives to behave honestly toward its depositors asymptotically to zero, even though every agent in the economy is risk neutral.

THREE MOST IMPORTANT INSIGHTS

First, delegating a task does not eliminate an incentive problem, it relocates it one layer up – any credible theory of intermediation must net the delegate's own agency cost against the gross benefit of delegation. Second, diversification can do incentive-relevant work even under risk neutrality, because its channel here is the probability of a costly triggering event (default on the deposit contract), not risk-sharing in the usual Arrow-Pratt sense. Third, the same functional form – debt with a bankruptcy-style penalty – emerges as optimal at both the borrower-to-intermediary and intermediary-to-depositor layers, giving a unified account of why both bank assets and bank liabilities look like debt.

STRONGEST EVIDENCE

The formal proof of Proposition 2, deriving $D_N \rightarrow 0$ directly from the weak law of large numbers applied to the shortfall probability P_N , with an explicit, checkable construction of the payment schedule $g_1(\cdot)$ and face value H_N that achieves it.

BIGGEST WEAKNESS

The central result is derived under independence (or a hedgeable factor structure); the paper does not characterize how quickly the delegation-cost advantage deteriorates as correlation rises, which is precisely the regime (systemic stress) in which the question matters most for financial stability.

MOST IMPORTANT RESEARCH GAP

No existing extension of the model jointly endogenizes intermediary capital structure, correlation-driven delegation cost, and Diamond-Dybvig-style run risk within a single dynamic framework, leaving the model's own implausibly low asymptotic capital buffer ($H_N \rightarrow N(R+K)$) without an internal resolution.

BEST RESEARCH OPPORTUNITY

Testing the model's own acknowledged boundary condition directly: whether banks with more correlated pre-crisis loan books experienced disproportionately larger funding-cost increases during 2008 and 2020, which would confirm that Proposition 2's diversification advantage empirically switches off exactly where the model predicts it should (Idea 2, Section 10).

THREE PAPERS TO READ NEXT

Townsend (1979); Diamond and Dybvig (1983); Holmström and Tirole (1997).

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